Soundness: provable implies valid

The Open Logic Project contributors

openlogic:propositional-logic/syntax-and-semantics:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-propositional-logic-syntax-and-semantics/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof argues both base cases and names the closure fact for the inductive step; the cited prop:semanticalfacts is an Exercise in the source and is not a counted card; the dependency is recorded as a prerequisite.

Editorial note. The label thm:soundness names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Allowed standard prerequisites (editorial).

- Semantic facts for entailment (modus-ponens closure). Same unit, \ol-label{prop:semanticalfacts} (source line 1101; Exercise in source). Line 1174-1175: the inductive step for modus ponens reads off the semantic consequence definition.

Source reference. Original proposition semanticalfacts: sources/olp-propositional-logic-syntax-and-semantics/pinned-original-source.tex, source lines 1101–1101. Prerequisite pointer invoked inside the counted proof (third item, modus-ponens closure of semantic consequence); it names a proposition of the same unit whose statement is used as a standard fact, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1 (Soundness)**.** *If* $\Gamma \vdash A$ *then* $\Gamma \models A$*.*

Proof. By induction on theorems. If A is an axiom then $\models A$ and hence a fortiori $\Gamma \models A$. Similarly if $A \in \Gamma$, $\Gamma \models A$. For the inductive step, suppose A is obtained by *modus ponens* from C and $C \to A$; also assume as inductive hypothesis that the theorem holds for $C \to A$ and C. The third item in semanticalfacts gives the desired result. □